

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

June 2017

EC 309.01 Digital Communication

Date: 01.06.2017, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

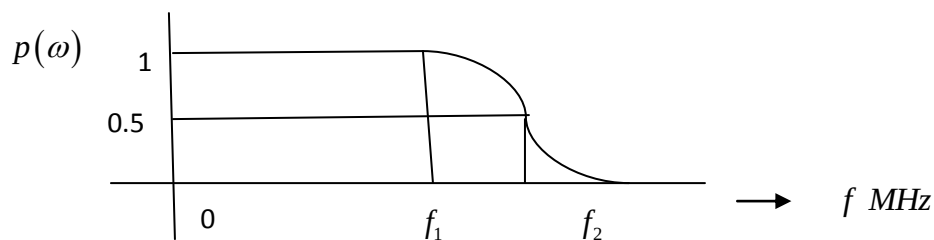
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

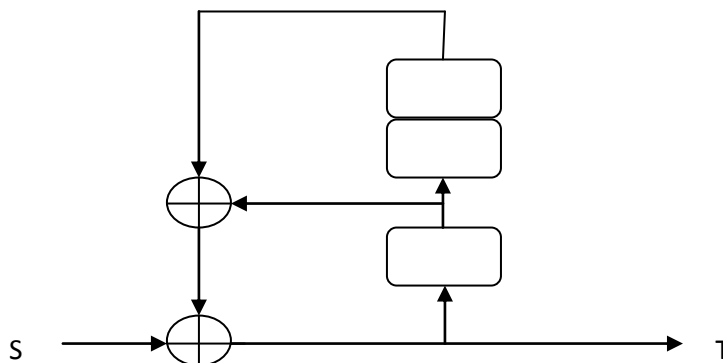
Q-1 A signal band limited to 1 MHz is sampled at a rate 50 % higher than the Nyquist rate [03]

(a) and quantized into 256 levels using a μ - law quantizer with $\mu = 255$. Determine the signal – to – quantization – noise ratio.

(b) Binary data at a rate of 1 Mbit/s is to be transmitted using Nyquist criterion pulse with [02]
 $p(\omega)$ shown in figure. The frequencies f_1 and f_2 (in hertz) of this spectrum are adjustable. The channel available for transmission of this data has a bandwidth of 700 kHz. Determine f_1 , f_2 and roll – off factor.



(c) A scrambler is shown in figure. If the sequence $S = 101010100000111$ is applied to the input of this scrambler determine the output sequence T. [04]



(d) Two event A and B are independent if and only if, [01]

(a) $P(A \cup B) = P(A) + P(B) - P(A)P(B)$; (b) $P(A/B) = P(B)$;

(c) $P(A \cap B)/P(A) = P(A/B)$; (d) All of the above

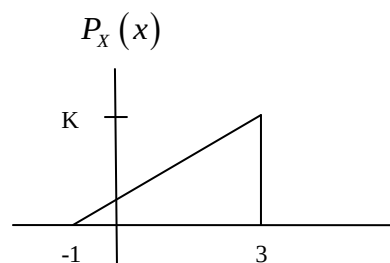
(e) For two event A,B it is always true that, [01]

(a) $P(A) + P(B) \geq 1$; (b) $P(A) + P(B) \leq 1$; (c) $P(A) - P(A \cup B) \leq P(A \cap B)$;

(d) $P(A/B) \leq P(B)$

Q-2 Attempt any two [12]

(a) Find the mean, the mean square, and the variance of the RV x whose PDF is shown in figure. [06]



(b) A transmitter has an alphabet of four letters $[x_1 x_2 x_3 x_4]$ and the receiver has an alphabet of three letters $[y_1 y_2 y_3]$. The joint probability matrix is, [06]

$$P(X,Y) = \begin{bmatrix} 0.3 & 0.05 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0.15 & 0.05 \\ 0 & 0.05 & 0.15 \end{bmatrix}$$

Calculate $H(X)$, $H(Y)$, $H(X,Y)$, $H(X/Y)$ and $H(Y/X)$

(c) A transmitter has an alphabet of two letters $[x_1 x_2]$ and the receiver has an alphabet of two letters $[y_1 y_2]$. The channel matrix is, $p(x_1) = 0.6$ and $p(x_2) = 0.4$ [06]

$$P(Y/X) = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$$

Calculate $I(X;Y)$

- Q-3 Attempt any two [12]**
- (a) In certain telemetry system, there are eight analog instruments, each of bandwidth 2 kHz. Sample of these signals are time-division multiplexed, quantized, and binary coded. The error in sample amplitudes cannot be greater than 1% of the peak amplitude. [06]
1. Determine L , the number of quantization levels.
 2. Find the transmission bandwidth B_T if nyquist criterion pulses with roll off factor $r = 0.2$ are used. The sampling rate must be at least 25% above the Nyquist rate.
- (b) Explain the generation of coherent binary phase shift keying (BPSK) modulation technique. Draw the constellation diagram for same. [06]
- (c) The level of M-ary PAM constellations with $M=8$ are $\{-7\alpha, -5\alpha, -3\alpha, -\alpha, \alpha, 3\alpha, 5\alpha, 7\alpha\}$. The probabilities of these symbol are related as $P((2i-1)\alpha) = 2P((2i-3)\alpha)$, for $-2 \leq i \leq 4$. The probability of $P(3\alpha)$ is? Determine $F_X(x)$. [06]

SECTION - II

- Q-4 In a random experiment, a trial consists of four successive tosses of a coin. If we define [03]**
- (a) an RV X as the number of heads appearing in a trial, determine $P_X(x)$ and $F_X(x)$. [03]
- (b) A Zero memory source emits six messages with probabilities 0.30, 0.25, 0.15, 0.12, 0.10, and 0.08 respectively. Find the 4-ary Huffman code. Determine its average word length, efficiency and redundancy of the code. [06]
- (C) A leased telephone of bandwidth 3 kHz is used to transmit binary data. Calculate the data rate (in bits per second) that can be transmitted if we use polar signal using nyquist criterion pulses of $r = 0.25$. [02]
- Q-5 Attempt any two [12]**
- (a) Explain the ISI and signal degradation study using eye diagram. [06]
- (b) A compact disc (CD) records audio signals digitally by using PCM. Assume the audio signal bandwidth to be 15 kHz. [06]
1. What is the Nyquist rate?
 2. If the Nyquist samples are quantized into $L = 65,536$ levels and then binary coded, determine the number of binary digits required to encode a sample.
 3. Determine the number of binary digit per second (bits/s) required to encode the audio signal.

- (c) In a Binary data transmission using duobinary pulses, sample values of the received pulses were read as follows. (First bit in the sequence is one) [06]

1 2 0 -2 -2 0 0 -2 0 2 0 0 2 0 0 0 -2

1. Explain if there is any error in detection
2. If there is no detection error, determine the received bit sequence.

Q-6 Attempt any two [12]

- (a) A source emits three equiprobable messages randomly and independently. [06]

1. Find the source entropy.
2. Find a compact ternary code, the average length of the code word, the code efficiency and the redundancy.

- (b) A box contains nine bad microchips. A good microchip is thrown into the box by mistake. Someone is trying to retrieve the good chip. He draws a chip randomly and tests it. If the chip is bad, he throws it out and draws another chip randomly, repeating the procedure until he finds the good chip. [06]

1. What is the probability that he will find the good chip in the first trial?
2. What is the probability that he will find the good chip in five trials?

- (c) Prove: SNR can be controlled by transmission bandwidth. [06]
